

Governance-Invariant MDPs: A Framework and Formal Safety Case for Agentic Wildfire Monitoring

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Abstract

Wildfire early warning must be fast enough to save lives yet accountable enough to trust: a false public alert can trigger a needless mass evacuation, while a missed detection can be fatal, so the authority to issue an alert cannot be ceded to an unaccountable autonomous agent. Constrained reinforcement learning (RL), the standard tool for safe autonomy, enforces safety only in expectation and thus permits violations. We introduce the *Governance-Invariant MDP* (GOMDP), which makes mandatory human authorization a hard, auditable, non-repudiable constraint enforced at the environment level as a cryptographic state-transition invariant rather than a policy penalty. Instantiated for wildfire monitoring with a Hyperledger Fabric smart contract and evaluated on three real fire events derived from NASA VIIRS/FIRMS data, GOMDP lets no public alert issue without a valid human authorization: under stated assumptions, any policy, however suboptimal or adversarial, satisfies the governance predicate with probability negligibly close to one (Theorem 1), and a closed-form breach bound (Theorem 2) delimits when Byzantine consensus outperforms a single verifier. A layered decomposition isolates what each enforcement layer buys. Across 20 seeds, PPO-GOMDP cuts detection latency by 17.5% versus a training-free baseline ($18.3 \rightarrow 15.1$ steps) at 100% governance compliance and cuts false public alerts from 22.4% to 6.0%, with statistically equivalent latency to constrained RL (TOST, $p = 0.004$) and robustness under adversarial stress. GOMDP is a deployable, auditable accountability layer for safety-critical civic AI.

1 Introduction

Wildfire early warning is shifting from human monitoring toward autonomous, agentic AI that fuses satellite, UAV, and sensor data to decide when to raise a public alarm. The stakes are asymmetric: a false alert can trigger a costly, dangerous mass evacuation, while a missed detection can cost lives, so the decision to broadcast must stay accountable to a human authority and auditable. Constrained MDPs (Altman 1999) and trust-region methods such as CPO (Achiam et al. 2017) enforce constraints only in expectation via Lagrangian relaxation, so violations remain common (5–15%

even after convergence (Achiam et al. 2017)); projection-based updates (Yang et al. 2020), distributional critics (Yang et al. 2021), and safety layers (Dalal et al. 2018) reduce but cannot eliminate them, since all tie their guarantee to policy or model quality (Gu et al. 2022). For public warning, an in-expectation, unaccountable guarantee is insufficient.

We address this gap with the *Governance-Invariant MDP* (GOMDP), in which the safety constraint is embedded in the environment’s transition function via cryptographic enforcement rather than in the policy’s reward signal. The structural insight is a separation of concerns: *safety is an environment property, not a policy property*. Any policy (random, greedy, or fully trained) satisfies the governance constraint with probability negligibly close to one, because the constraint is a cryptographic state-transition invariant no action can bypass. The guarantee is scoped: it secures *authorization integrity and accountability* (no alert without a valid, non-repudiable human authorization), not the correctness of the human decision, which no cryptography can supply. The formal results identify the precise assumptions under which this guarantee holds; the experiments carry the paper’s primary evidentiary weight.

We instantiate the framework for wildfire early warning (Di Virgilio et al. 2019; Ghali, Akhloufi, and Mseddi 2022): a Hyperledger Fabric (Androulaki et al. 2018) smart contract refuses to execute the alert-broadcast transition unless a cryptographically valid human authorization signature is present, turning human oversight from a procedural check into an environment constraint for which a fire agency’s duty officer, not the autonomous agent, remains the named authorizer, fitting existing emergency-operations authorization chains. It integrates multi-agent UAV coordination (a greedy policy and a deep PPO policy trained inside the GOMDP) with two-stage Bayesian anomaly verification.

Contributions. (C1) Safety–optimality decoupling for agentic RL: we formalize the GOMDP, an MDP whose safety constraint is a cryptographic environment-level invariant, and show (Corollary 1) that the safety certificate is independent of the policy’s optimality gap, unlike CMDP/CPO/WCSAC, whose violation probability scales with that gap. **(C2) A layered empirical methodology** that separates the marginal contribution of cryptographic authentication from that of Byzantine consensus (the Central+Sig ablation), absent from prior shielding and safety-layer evaluations, and that quan-

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tifies what governance costs: because the gate acts after detection, enforcement leaves detection latency unchanged and adds ≈ 4.2 steps to the detection-to-broadcast path, while cutting the false public alert rate from 22.4% to 6.0% at full compliance. **(C3)** *A formally scoped safety case*: Theorem 1 identifies the minimal assumption set under which any policy satisfies the governance predicate with probability negligibly close to one, and Theorem 2 gives a closed-form breach bound with an explicit envelope for when Byzantine consensus does and does not outperform a single signature-verifying server. **(C4)** *A broad evaluation* against constrained RL (PPO-CMDP, WCSAC), cryptography-free environment enforcement (logical shielding, SafeLayer), and a signature-only server, over three VIIRS-derived events and strategic as well as i.i.d. adversaries.

2 Related Work

Constrained RL and Safe AI Altman’s CMDP framework (Altman 1999) enforces constraints via Lagrangian relaxation, providing in-expectation guarantees; CPO (Achiam et al. 2017) and RCPO (Tessler, Mankowitz, and Mannor 2019) reduce but cannot eliminate violations, and PCPO (Yang et al. 2020) projects updates onto the constraint set without removing them. WCSAC (Yang et al. 2021) adds a distributional (CVaR-based) safety critic (90.6% compliance in our evaluation) but remains policy-level. SafeLayer (Dalal et al. 2018) projects actions per step via a learned linearized cost model, stronger than Lagrangian methods but inheriting model error (Section 6.2). Safe shielding (Alshiekh et al. 2018) filters unsafe actions at the environment boundary, architecturally closest to GOMDP; our shield baseline shows this enforcement is complete against the policy’s own actions yet defenseless against external injection. Surveys (Gu et al. 2022) confirm per-trajectory guarantees under adversarial conditions remain open for policy-level methods; GOMDP extends shielding with cryptographic enforcement and Byzantine-tolerant consensus, a threat model absent from prior shielding work.

Runtime Assurance, CBFs, and Monitoring The Simplex architecture (Sha 2001) pioneered environment-boundary enforcement in aerospace: a verified safety controller overrides an unverified performance controller when a safety envelope is threatened. GOMDP inherits this philosophy but replaces the trusted local monitor with a distributed cryptographic mechanism, changing the threat model from component failure to adversarial compromise. Control Barrier Functions (Ames et al. 2019) enforce forward-invariant safe sets for continuous dynamics; the GOMDP predicate is discrete and requires non-repudiation beyond CBF synthesis. Runtime monitoring (Leucker and Schallhart 2009) observes and reacts to events, whereas GOMDP embeds enforcement in the transition function itself; shield synthesis from LTL specifications (Alshiekh et al. 2018) is orthogonal to how the resulting predicate is enforced at runtime.

AI-Based Wildfire Detection Contemporary systems apply deep learning and multimodal fusion (Ghali, Akhloufi, and Mseddi 2022). IoT threshold pipelines (Tarigan, Robet, and Tarigan 2026) report mean detection latency near 45 time

units without adaptive patrol, consistent with our Static baseline (41.5 steps; context only). Partheepan et al. (Partheepan, Sanati, and Hassan 2023) identify governance and accountability as open challenges in autonomous UAV monitoring, the gap GOMDP addresses; constrained safe RL for autonomous driving (Gao et al. 2024) tackles a related domain with soft Lagrangian constraints.

Blockchain Accountability and Verified Contracts Permissioned blockchains such as Hyperledger Fabric enable authenticated, tamper-evident execution with Byzantine fault-tolerant (BFT) ordering, while prior blockchain-based disaster monitoring primarily uses distributed ledgers for post-hoc auditing rather than real-time governance enforcement. Assumption 1 connects our guarantee to the smart-contract verification literature (Bhargavan et al. 2016). Human-in-the-loop (HITL) and AI-oversight architectures reduce automation error but treat oversight as an advisory or interface-level check (Holzinger, Zatloukal, and Müller 2025); GOMDP instead makes oversight a hard, non-repudiable environment constraint no policy can bypass.

3 The Governance-Invariant MDP Framework

We introduce the GOMDP as a system design for environment-level safety enforcement, together with a formal security argument that makes explicit the assumptions under which the design’s central guarantee holds.

3.1 Formal Definition

Definition 1 (Governance-Invariant MDP). *A Governance-Invariant MDP (GOMDP) is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}_G, r, \Omega, O, \mathcal{G})$, where $\mathcal{S}, \mathcal{A}, r, \Omega, O$ are standard partially observable MDP (POMDP) components with per-step reward $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, $\mathcal{G} : \mathcal{S} \times \mathcal{A} \rightarrow \{0, 1\}$ is a governance predicate, and $\mathcal{T}_G$ is the governance-constrained transition function: $\mathcal{T}_G(s' | s, a) = \mathcal{T}(s' | s, a)$ if $\mathcal{G}(s, a) = 1$ or $a \notin \mathcal{A}_{\text{alert}}$, and $\mathcal{T}_G(\text{block} | s, a) = 1$ (the alert action is blocked) if $a \in \mathcal{A}_{\text{alert}}$ and $\mathcal{G}(s, a) = 0$. This makes $\mathcal{T}_G$ total over $\mathcal{S} \times \mathcal{A}$: the governance predicate gates only alert actions, and every non-alert action proceeds under the unconstrained kernel $\mathcal{T}$ regardless of $\mathcal{G}$. The enforcement of $\mathcal{G}$ is cryptographic: the environment refuses to execute alert state transitions lacking a cryptographically valid governance certificate, independently of the policy.*

The distinction from a CMDP is structural: a CMDP constraint modifies the learning signal via a Lagrange multiplier, so a sufficiently suboptimal or adversarial policy can still act unsafely, whereas a GOMDP environment cannot produce a state with $\text{Alert} = 1$ unless $\mathcal{G}(s, a) = 1$. This also separates GOMDP from logical shielding (Alshiekh et al. 2018), which lacks cryptographic non-repudiation and BFT consensus, and from SafeLayer (Dalal et al. 2018), whose learned projection carries model error. The structural separation, not a new cryptographic primitive, is the central claim; the formal result below pins down the assumptions it rests on.

Assumption 1 (Faithful Implementation). *The deployed smart-contract chaincode is a correct, bug-free realization*

of $\mathcal{T}_G$ as specified in Definition 1: for every (s, a) with $a \in \mathcal{A}_{\text{alert}}$, the chaincode evaluates $\mathcal{G}(s, a)$ and blocks the transition whenever $\mathcal{G}(s, a) = 0$. Formal program verification of the chaincode, for which established techniques exist (Bhargavan et al. 2016), is outside the scope of this paper; we treat this assumption explicitly and return to its consequences in the Limitations.

3.2 Policy-Agnostic Safety

Theorem 1 (Policy-Agnostic Safety). *Let $\Sigma = (\mathcal{G}\text{-MDP}, \Pi_{\text{Ed25519}}, \mathcal{V})$ be a GOMDP instantiation satisfying Assumption 1, where Π_{Ed25519} is a $(t_{\text{adv}}, \epsilon_{\text{sig}})$ -secure digital signature scheme (Bernstein et al. 2012) and $\mathcal{V}$ is a set of k validators of which at most f are Byzantine, with $k \geq 3f + 1$ (equivalently, $f \leq \lfloor (k - 1)/3 \rfloor$). Let π be any stochastic policy (including arbitrarily suboptimal or adversarially chosen policies) operating within Σ with governance predicate $\mathcal{G}(s, a) = [\text{Conf}_t^{(2)} > \tau \wedge \text{HA}_t = 1]$. Then, for any adversary running in time $t \leq t_{\text{adv}}$ and for all horizons T :*

$$P_\pi \left(\exists t \leq T : \text{Alert}_t = 1 \wedge (\text{Conf}_t^{(2)} \leq \tau \vee \text{HA}_t = 0) \right) \leq T \cdot \epsilon_{\text{sig}} \quad (1)$$

where $T \cdot \epsilon_{\text{sig}} \approx 0$ under standard hardness assumptions for Ed25519 ($\epsilon_{\text{sig}} \approx 2^{-128}$). Conditional on Assumption 1, the guarantee is independent of π and holds with probability negligibly close to one over all trajectories.

Proof Sketch. By reduction: an adversary producing an unauthorized alert must, under Assumption 1, either forge a valid certificate without an authorized validator key (Case 1) or corrupt more than f validators (Case 2, handled by Theorem 2). Case 1 yields an Ed25519 forger with the same success probability, bounded by ϵ_{sig} ; a union bound over T timesteps gives (1). The proof does not rule out a chaincode bug; that failure mode is excluded by hypothesis (Assumption 1), so (1) is a guarantee about the specification conditional on faithful implementation, not an unconditional guarantee about any deployed artifact. The full proof appears in the technical appendix. $\square$

Corollary 1 (Safety–Optimality Decoupling). *The guarantee of Theorem 1 holds for any policy π , including the worst-case policy that minimizes return; the optimality gap does not affect the safety certificate. This contrasts with CMDP/CPO/WCSAC, where constraint violation probability depends directly on the optimality gap during training (Achiam et al. 2017; Yang et al. 2021).*

Remark 1 (Novelty and Scope). *The novelty is scoped precisely. That an environment blocking unauthorized transitions by construction cannot be made to permit them is a definitional consequence of Definition 1; that secure signatures resist forgery is standard. Theorem 1 contributes the connective step: it identifies the minimal assumption set under which environment-level enforcement transfers a guarantee from specification to any policy, its reduction locates exactly where the guarantee would fail (broken primitive, validator supermajority, or implementation bug), and it distinguishes this guarantee from shielding (Alshiekh et al. 2018), which cannot make the same claim under an adversarial validator*

model. The result is a security argument for the system design, which, with its instantiation and evaluation, is the paper’s primary contribution.

3.3 Adversarial Robustness Bound

Theorem 2 (Adversarial Robustness). *Under the threat model of Section 4.2, with k validators, Byzantine tolerance $f \leq \lfloor (k - 1)/3 \rfloor$, and independent per-validator compromise probability p_c , the probability of a successful alert-injection breach in a GOMDP is bounded by the probability that more than f validators are compromised:*

$$P_{\text{break}}^{\text{GOMDP}} \leq \sum_{i=f+1}^k \binom{k}{i} p_c^i (1 - p_c)^{k-i}. \quad (2)$$

For a centralized signature-verifying server, breach requires compromising the single verifier: $P_{\text{break}}^{\text{sig}} = p_c$. For a centralized system without authentication, breach requires only channel access: $P_{\text{break}}^{\text{central}} = p_{\text{attack}}$.

Proof Sketch. A breach requires either certificate forgery (probability $\leq T \epsilon_{\text{sig}} \approx 0$ by Theorem 1) or malicious consensus, which needs more than f compromised validators; the binomial tail follows from independence. The signature-only server has a single verification point, so its breach probability under key compromise is exactly p_c . Without authentication, a single compromised channel suffices. $\square$

Numerical characterization. For $k = 7, f = 2$: $P_{\text{break}}^{\text{GOMDP}} \leq 0.004, 0.026, 0.148, 0.353$ at $p_c = 0.05, 0.10, 0.20, 0.30$ (exact binomial tails of Eq. (2)). Two observations: $p_c = 0.30$ is a pessimistic stress point, not a defensible operating regime, and the bound reflects this; and against the single signature verifier ($P_{\text{break}}^{\text{sig}} = p_c$), the $k = 7$ tail is below p_c only for $p_c \lesssim 0.26$, above which a single hardened verifier is numerically preferable and the case for consensus rests on non-repudiation and the absence of a single point of failure. Section 6.5 tests this decomposition empirically.

4 System Model and Threat Assumptions

The wildfire monitoring architecture instantiates the GOMDP framework across five layers, shown in Fig. 1: physical IoT sensing, digital twin, agentic coordination, blockchain governance, and human validation (a principal-notation summary accompanies the supplementary technical appendix).

4.1 System Architecture

Figure 1 details the pipeline. A heterogeneous sensor ensemble (UAV thermal cameras with detection probability 0.85, ground IoT sensors, GOES-16 feeds, meteorological stations) generates observations fused probabilistically (Khaleghi et al. 2013); in the VIIRS-data evaluation (Section 6.3) this layer is replaced by preprocessed VIIRS 375-m active fire data, and no physical UAVs or blockchain instances were deployed in any experiment. A wildfire-risk digital twin drives three agents (fire-risk prediction, UAV coordination via Algorithm 1 or PPO-GOMDP, Section 5.2, and two-stage anomaly verification), with EMA-based threshold adaptation between

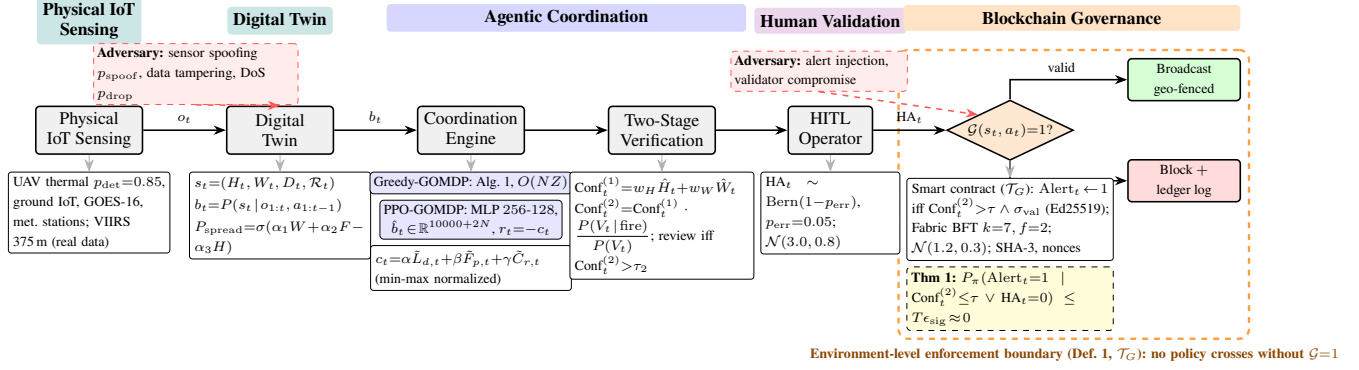


Figure 1: System architecture instantiating the GOMDP framework (Def. 1) across the five layers of Section 4. The Fabric smart contract enforces the governance predicate $\mathcal{G}$ (4) as a cryptographic state-transition invariant $\mathcal{T}_G$ (BFT, $k=7, f=2$), so every policy satisfies Theorem 1. Red panels: threat model (Section 4.2). No physical UAVs or blockchain were deployed.

episodes. Anomaly events are committed as Hyperledger Fabric (Androulaki et al. 2018) transactions under BFT ordering ($k = 7, f = 2, k \geq 3f + 1$); the smart contract enforces the governance predicate (4), atomically checking the confidence threshold and Ed25519 validator signature before alert dissemination. Operators review via an explainable-AI (XAI) dashboard (response time $\mathcal{N}(3.0, 0.8)$ steps); a multi-signature extension targets single-operator oracle risk (Section 6.4), and the experiments use the single-operator model of Section 5.4.

4.2 Threat Model

We consider five adversarial capabilities (Fig. 1): **sensor spoofing** (fabricated heat readings with probability p_{spoof} , i.i.d. or targeted at high-belief cells (Shafique, Mehmood, and Elhadef 2021)); **data tampering** (Huang, Liu, and Bi 2021); **alert injection** (a public alert bypassing the governance pipeline (He et al. 2015)); **validator compromise** (Byzantine control of consensus); and **denial-of-service** (packet drops p_{drop} (Duo, Zhou, and Abusorrah 2022)). Section 6.5 reports breach rates; Theorem 2 bounds injection and compromise.

Human Oracle Risk GOMDP secures *authorization integrity* (no alert without a valid, attributable human signature), not *decision correctness*: a validator may authorize incorrectly, and on-chain logging makes such decisions non-repudiable and auditable but does not prevent them. We report robustness to the operator error rate p_{err} up to 0.20 (appendix), and an m -of- n multi-signature quorum (Section 6.4) reduces single-operator risk; correlated fatigue and adversarial social engineering remain outside the model. The HITL false-negative rate is in Table 1.

5 Governance-Constrained Agentic Architecture

5.1 GOMDP Instantiation for Wildfire Monitoring

The wildfire monitoring GOMDP has state $s_t = (H_t, W_t, D_t, \mathcal{R}_t)$ and belief state $b_t = P(s_t | o_{1:t}, a_{1:t-1})$ updated via Bayesian filtering. The optimization objective is

a scalar cost to be *minimized*:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=1}^T c_t \right], \quad c_t = \alpha \tilde{L}_{d,t} + \beta \tilde{F}_{p,t} + \gamma \tilde{C}_{r,t}, \quad (3)$$

with $\alpha + \beta + \gamma = 1$, where L_d is detection latency (steps), F_p is the false public alert rate (%), and $C_r = \sum_{u \in \mathcal{U}} d(u, p_u^*)/N$ is the normalized UAV repositioning cost. Each per-step contribution is min-max normalized to $\tilde{X}_t \in [0, 1]$ using the running minimum and maximum over the training trajectory, before weights $(\alpha, \beta, \gamma) = (0.50, 0.35, 0.15)$ are applied; the PPO agent maximizes $r_t = -c_t$ (scaled to $[-1, 1]$; Section 5.2). Running-extrema normalization makes the reward mildly non-stationary early in training, but extrema stabilize within 50 episodes with no attributable divergence (appendix). The governance predicate is:

$$\mathcal{G}(s_t, a_t) = 1 \left[\text{Conf}_t^{(2)} > \tau \wedge HA_t = 1 \right], \quad (4)$$

enforced cryptographically by the smart contract (8), with $\tau = \tau_2 = 0.80$ throughout. By Theorem 1, any policy optimizing (3) satisfies (4) with probability negligibly close to one.

5.2 PPO-GOMDP: Deep RL Policy

To demonstrate that GOMDP accommodates arbitrary policies while preserving safety, we train a PPO (Schulman et al. 2017) agent *inside* the GOMDP environment. The governance constraint is *not* in the reward signal; it is enforced by the transition function, and Theorem 1 holds even before the agent learns to avoid non-compliant alert attempts.

Policy Architecture The policy is a two-layer MLP (hidden dimensions 256 and 128, ReLU) mapping a normalized, flattened belief-state vector $\hat{b}_t \in \mathbb{R}^{10000+2N}$ (min-max normalized 100×100 belief map concatenated with normalized UAV positions) to sector-assignment logits per UAV. Alert request generation is handled by the verification pipeline (Section 5.4), not the policy: the policy controls *where* to look, the governance predicate controls *whether* to alert. The flattened-MLP design reflects the input’s nature (a Bayesian posterior map, not a raw image, with weak local-correlation structure) and the small discrete action space; we did not

evaluate a convolutional encoder, and whether one would improve sample efficiency on this input remains open.

Training Training proceeds for 1000 episodes on the 100×100 grid with learning rate 3×10^{-4} , clip ratio $\epsilon = 0.2$, entropy bonus 0.01, discount $\gamma_{\text{RL}} = 0.99$, 4 update epochs per episode, batch size 64. Reward is scaled to $[-1, 1]$ by dividing by the maximum single-episode return over the first 50 training episodes. Training terminates when validation L_d fails to improve by > 0.5 steps over 100 consecutive episodes (typically by episode 750, with the validation curve plateauing at $L_d \approx 15.1$ by episode 650; the per-episode validation curve accompanies the code package).

5.3 Greedy Approximation Policy

The risk-weighted greedy policy (Algorithm 1) sorts sectors by belief-derived risk and assigns UAVs by proximity within energy budgets, at $O(NZ)$ per step; it provides a tractable, training-free baseline within the same GOMDP environment.

Algorithm 1 Risk-Weighted Greedy Coordination (Greedy-GOMDP)

Require: Belief state b_t ; fleet $\mathcal{U}$; risk map $\mathcal{R}_t$; sector set $\mathcal{P}$ of size $Z = |\mathcal{P}|$ ($Z = 25$ for a 5×5 sector partition of the 100×100 grid)

Ensure: Redeployment commands $\{(u_i, p_i)\}$

- 1: Compute $\{\mathcal{R}_t^{p_i}\}$ from b_t ; sort sectors descending by $\mathcal{R}_t^{p_i}$
- 2: **for** each $u_i \in \mathcal{U}$ (sorted by proximity) **do**
- 3: Assign u_i to highest-unassigned sector p^* within energy budget; issue (u_i, p^*)
- 4: **end for**
- 5: **return** redeployment commands $\triangleright O(NZ)$ per step

5.4 Two-Stage Probabilistic Verification

Stage-1 cross-modal fusion implements a weighted linear combination:

$$\text{Conf}_t^{(1)} = w_H \hat{H}_t + w_W \hat{W}_t, \quad w_H = 0.65, \quad w_W = 0.35. \quad (5)$$

Stage-2 applies a Bayesian update upon secondary UAV verification:

$$\text{Conf}_t^{(2)} = \text{Conf}_t^{(1)} \cdot \frac{P(V_t | \text{fire})}{P(V_t)}, \quad (6)$$

where $P(V_t | \text{fire}) = 0.85$ and $P(V_t)$ marginalizes over $\text{Conf}_t^{(1)}$ as a fire prior. Both thresholds τ_1 and $\tau_2 = \tau$ are adapted online via EMA of episode-level precision-recall. Human review is triggered if and only if $\text{Conf}_t^{(2)} > \tau_2$; the governance predicate (4) is evaluated subsequently by the smart contract.

HITL Simulation Methodology Because review is triggered only for events with $\text{Conf}_t^{(2)} > \tau_2$, every event an operator sees has already cleared the confidence threshold; we therefore model the operator’s decision as a *constant* approval probability over the reviewed population:

$$\text{HA}_t \sim \text{Bernoulli}(1 - p_{\text{err}}), \quad p_{\text{err}} = 0.05, \quad (7)$$

where p_{err} captures workload- and fatigue-driven baseline rejection, consistent with rejection rates in HITL alarm management studies (Holzinger, Zatloukal, and Müller 2025); a confidence-dependent approval curve would be evaluated only on its flat tail here, so a constant-rate model is the appropriate first-order approximation absent human-subject calibration data. $FN_r = 2.1\%$ (Table 1) aggregates stage-1, stage-2, and HITL rejection. Sensitivity to $p_{\text{err}} \in \{0.05, \dots, 0.20\}$ is in the appendix; a genuinely confidence-calibrated operator model requires human-subject data and remains future work. Response delay is $\mathcal{N}(3.0, 0.8)$ steps, truncated at > 0 (Holzinger, Zatloukal, and Müller 2025).

5.5 Blockchain-Enforced Governance Invariant

The smart contract implements $\mathcal{T}_G$ for alert actions:

$$\text{Alert}_t \leftarrow 1 \text{ iff } \text{Conf}_t^{(2)} > \tau \wedge \sigma_{\text{validator}} \text{ is valid (Ed25519),} \quad (8)$$

verified atomically by the Fabric chaincode. Approved alerts generate a geo-fenced payload over redundant channels; all events are committed to the immutable ledger with SHA-3 hashed evidence for non-repudiation, and per-event nonces prevent replays.

6 Experimental Evaluation

We evaluate across four regimes: (A) simulation benchmarking with policy comparison; (B) VIIRS-data simulation evaluation; (C) ablation; (D) adversarial robustness and stress testing.

6.1 Setup

Simulation Environment Wildfire dynamics on a 100×100 grid; stochastic cellular automaton propagation $P_{\text{spread}} = \sigma(\alpha_1 W + \alpha_2 F - \alpha_3 H)$ with weights (0.45, 0.35, 0.30), the calibration used in all reported runs, tuned to reproduce spread rates of 1–4 cells per 10-step window; $T = 3000$ steps (10 s each); 20 seeds (0–19; per-seed metric files in the code package). The full parameter table is in the appendix.

VIIRS-Data Simulation Three Visible Infrared Imaging Radiometer Suite (VIIRS) 375-m active fire events were preprocessed into 100×100 grids: California August Complex 2020, Mediterranean (Evia) 2021, and NSW Black Summer 2019–20 (bounding boxes in the code package); hotspots came from the NASA FIRMS API with ERA5 forcing. These experiments run the GOMDP pipeline on real fire-distribution inputs; no physical UAV or blockchain instance was used, and latency is in simulation timesteps.

Configurations Nine configurations. Governed: **PPO-GOMDP** (deep RL); **Greedy-GOMDP** (Algorithm 1). Environment-level enforcement without cryptography: **Shield-PPO** (the same predicate enforced by a local logical shield after Alshiekh et al. (2018), no signatures or consensus); **SafeLayer** (Dalal et al. 2018) (per-step action projection, learned linearized cost model). Authentication without consensus: **Central+Sig** (the PPO-GOMDP pipeline with BFT consensus replaced by one centralized Ed25519-verifying server). Policy-level surrogates (see Statistical Test-

ing): **PPO-CMDP** (Lagrangian-style soft constraint); **WC-SAC** (Yang et al. 2021) (same action space and safety budget). Ungoverned: **Adaptive AI** (greedy); **Static** (fixed patrol). **Adaptive AI** uses the identical coordination (Algorithm 1) and two-stage verification as **Greedy-GOMDP** but is never gated by $\mathcal{G}$, isolating the marginal effect of the governance gate alone.

Statistical Testing Paired two-sided t -tests with Holm-Bonferroni correction; all $p < 0.01$ unless noted. Compliance is episode-level (fraction of episodes with zero violations); it is definitional for the GOMDP rows (Theorem 1) and for Adaptive AI/Static (0%), so neither is an independent measurement. The policy-level rows (PPO-CMDP, WCSAC, SafeLayer) are *surrogates* reproducing the enforcement semantics of Lagrangian and projection methods – a soft constraint admitting residual violations – not reimplementations; their compliance characterizes in-expectation enforcement in our environment, not those specific published methods, and lands in the 5–15% residual-violation regime CPO reports (Achiam et al. 2017).

Compute and Code Availability Training one PPO-GOMDP policy to convergence (≈ 650 episodes, ≈ 2.6 M parameters) took ≈ 2.5 GPU-hours on a consumer-grade GPU (PPO-CMDP and WCSAC comparable); cumulative budget ≈ 110 GPU-hours; inference is under 5 ms on CPU. Code, configurations, preprocessing scripts, and the per-seed metric files are provided as a supplementary code and data archive with this submission.

6.2 Simulation Benchmarking and Policy Comparison

Table 1 reports the key finding: PPO-GOMDP reduces detection latency by 17.5% relative to Greedy-GOMDP ($18.3 \rightarrow 15.1$ steps) while maintaining 100% governance compliance. PPO-CMDP achieves comparable latency (14.8 steps) but violates the governance predicate in 7.2% of evaluation episodes; WCSAC achieves $L_d = 14.6$ steps but violates in 9.4%. SafeLayer’s learned projection cuts violations to 1.6% of episodes, materially better than the Lagrangian methods but short of a guarantee, and Shield-PPO matches GOMDP compliance in the benign setting while offering no defense against external injection (Section 6.5).

A paired t -test on L_d (PPO-GOMDP vs. PPO-CMDP) yields $t(19) = 1.28$, two-sided $p = 0.216$ (still non-significant after Holm-Bonferroni, which can only increase p); non-significance does not itself establish equivalence. We therefore apply a two one-sided tests (TOST) procedure (Lakens 2017) with a pre-specified margin $\Delta = 1.0$ step (10 s wall-clock, negligible against a 20–40 minute pre-ignition window): with observed paired difference 0.3 steps (SE = 0.234), $t_L(19) = 5.55$ ($p < 0.001$) and $t_U(19) = -2.99$ ($p = 0.004$), so $\max(p_L, p_U) = 0.004 < 0.05$ confirms equivalence within Δ . Because the governance gate acts only after detection (Section 7), no latency difference between governed and ungoverned coordination is expected from the mechanism itself; any observed difference reflects the coordination policy, not the enforcement layer. The governance invariant is verified per episode by checking whether any alert was broadcast without a valid on-chain human authorization,

Table 1: Policy comparison ($N=20$, $n=20$ seeds; mean $\pm$ std). FN_r : missed-detection rate. ^{def}: compliance is definitional (Theorem 1 for governed rows; no gate for ungoverned rows), not independently measured.

Method	L_d	F_p (%)	FN_r (%)	Compl.	Enforce.
PPO-GOMDP	15.1 $\pm$ 1.1	6.0 $\pm$ 1.1	2.1 $\pm$ 0.9	100.0% ^{def}	Crypto
Greedy-GOMDP	18.3 $\pm$ 1.4	6.1 $\pm$ 1.3	2.3 $\pm$ 1.0	100.0% ^{def}	Crypto
Central+Sig	15.0 $\pm$ 1.1	6.0 $\pm$ 1.2	—	100.0% ^{def}	Sig. only
Shield-PPO	15.2 $\pm$ 1.2	6.2 $\pm$ 1.3	—	100.0%	Logical
SafeLayer	14.9 $\pm$ 1.1	7.0 $\pm$ 1.6	—	98.4%	Learned
PPO-CMDP	14.8 $\pm$ 1.0	8.3 $\pm$ 2.4	2.6 $\pm$ 1.0	92.8%	Lagrangian
WCSAC	14.6 $\pm$ 1.2	9.4 $\pm$ 2.0	3.8 $\pm$ 1.4	90.6%	Lagrangian
Adaptive AI	16.2 $\pm$ 1.2	22.4 $\pm$ 2.1	0.9 $\pm$ 0.5	0.0% ^{def}	None
Static	41.5 $\pm$ 3.1	15.3 $\pm$ 2.4	1.8 $\pm$ 0.8	0.0% ^{def}	None

and injection resistance is measured by mounting forged and replayed authorization attempts against the live contract rather than assumed (Section 6.4).

Fleet-size-dependent trends (full plots in the appendix) show all governed and Lagrangian configurations converging toward similar per-UAV latency as N grows from 5 to 40, while Static and Adaptive AI retain a persistent gap; the ranking in Table 1 is stable across $N \in \{5, 10, 20, 40\}$, and the appendix reports the full metric breakdown (including blockchain and human-review latency and governance overhead) for all nine configurations.

Scope of Baselines Our policy-level comparators are Lagrangian and projection methods (PPO-CMDP, WCSAC, SafeLayer). More recent constrained-RL methods such as SafeDreamer (Huang et al. 2024) and CCPO (Yao et al. 2023) remain policy-level and so inherit the same in-expectation limitation, but we did not implement them; a controlled comparison against them is future work.

6.3 VIIRS-Data Simulation Evaluation

Performance on three VIIRS-derived fire events (full per-event table in the supplementary technical appendix), with governed and constrained baselines on every event so the evaluation is a comparison rather than a showcase, confirms the pattern transfers: latency exceeds the pure-synthetic setting (21.8–24.1 steps vs. 15.1 for PPO-GOMDP) due to real sensor noise and geographic complexity, and false alert rates are elevated (7.9–9.1% vs. 6.0%) due to non-fire heat sources; the governance invariant holds in all episodes, and PPO-CMDP’s compliance deficit persists (92.4–93.6%), so the qualitative ranking transfers. We do not report the Tarigan et al. (Tarigan, Robet, and Tarigan 2026) IoT pipeline as a controlled improvement, since it differs in hardware, scenarios, modality, and latency definition; its ≈ 45 -unit figure is qualitative context only. Adaptive AI reaches lower latency but 23.9–26.1% false alert rates, confirming that unrestrained alert dissemination is operationally hazardous under realistic noise.

6.4 Ablation Study

Table 2 isolates each component and, critically, separates *authentication* from *consensus*. Injection resistance is *measured*, not *asserted*: each attempt constructs a forged authorization and submits it through the same contract entry point a legitimate alert traverses, in three variants – an unsigned certificate,

Table 2: Ablation ($N=20$, 20 seeds). Inj.: injection attacks blocked (of 100). “—Consensus” keeps Ed25519 verification at one server; “—All authentication” removes signatures and consensus.

Configuration	L_d	F_p (%)	Inj.
PPO-GOMDP (full)	15.1 ± 1.1	6.0 ± 1.1	100/100
Greedy-GOMDP (full)	18.3 ± 1.4	6.1 ± 1.3	100/100
– Adaptive coordination	29.7 ± 2.6	6.1 ± 1.2	100/100
– HITL authorization	15.2 ± 1.1	22.4 ± 2.2	100/100
– Consensus (Central+Sig)	15.0 ± 1.1	6.0 ± 1.2	100/100
– All authentication	15.1 ± 1.1	6.9 ± 1.4	0/100
– Multi-stage verif.	15.0 ± 1.1	14.8 ± 2.0	100/100
PPO-CMDP (no blockchain)	14.8 ± 1.0	8.3 ± 2.4	0/100

a cryptographically valid signature from a non-validator key, and a replay of a previously approved certificate – so a blocked attempt reflects the enforcement mechanism rejecting real input. “— Adaptive coordination” replaces risk-weighted ranking with round-robin while keeping governance intact (distinct from Static, which has no governance), isolating the ranking’s latency value; removing HITL authorization isolates its contribution to F_p . The two authentication rows separate what a single “blockchain removed” ablation would confound: “— Consensus” (Central+Sig) still blocks all 100 injection attempts, so *signature verification alone* defeats direct injection, and only “— All authentication” admits injections; the consensus layer’s distinct contributions are therefore validator-compromise resilience and non-repudiation (Theorem 2), not injection resistance per se. A lightweight m -of- n multi-signature quorum (Section 4.1) is evaluated in the appendix.

6.5 Adversarial Robustness and Stress Testing

We compare three architectures under attack: GOMDP, the centralized signature-verifying server (Central+Sig), and the unauthenticated centralized system; validator-compromise figures are summarized below and tabulated in full in the supplementary technical appendix, together with the full per-attack spoofing and injection breakdown.

Attack Resistance Sensor spoofing increases F_p for all architectures (it bypasses governance and corrupts the belief state directly), but the HITL gate bounds the damage (i.i.d. $p = 0.20$: GOMDP $F_p = 9.4\%$ vs. unauthenticated 38.7% ; targeting high-belief cells raises this only to 8.6% at $p = 0.10$), and compliance stays 100% throughout since spoofing cannot manufacture a valid authorization certificate. Direct alert injection is blocked completely by GOMDP and Central+Sig and succeeds completely against the unauthenticated system, so injection resistance is correctly attributed to *Ed25519 signature verification* rather than the blockchain layer generally: an unauthenticated baseline conflates signatures and consensus, while Central+Sig isolates signatures as the operative mechanism. Under *validator compromise*, stochastic and deterministic-targeted models behave differently: consensus helps below the $p_c \approx 0.26$ crossover (Theorem 2), and deterministic compromise requires $f_c \geq f+1 = 3$ validators for breach, confirming the tolerance boundary ($f_c = 2$ tolerated, $f_c = 3$ not, F_p rising to 8.9%). A validator-count sweep appears in the appendix; the governance invariant holds

throughout.

7 Discussion

Why GOMDP, and What Each Layer Buys Theorem 1 and the layered decomposition (Tables 1–2) attribute each layer’s role: signature verification alone defeats direct alert injection; BFT consensus adds validator-compromise resilience below the $p_c \approx 0.26$ crossover plus non-repudiation; and the logical shield matches GOMDP compliance against the policy’s own actions but, lacking authentication, is defenseless against external injection.

What Governance Costs The governance gate is evaluated *after* detection and verification, so by construction it cannot change detection latency L_d : a policy finds a fire exactly as fast whether or not its alert will subsequently require authorization. What governance costs is the delay between detection and public broadcast, $\Delta_{BC} + \Delta_{HITL} \approx 1.2 + 3.0 = 4.2$ steps (≈ 42 s at 10 s per step). Against a 20–40 minute pre-ignition window this is a small price for a non-repudiable authorization record, and it is the honest accounting: the benefit of governance appears in F_p (Table 1), not in L_d . We note this because an apparent latency *improvement* from adding governance would indicate a confound rather than a mechanism.

When Is Consensus Worth It, and Does It Generalize?

At realistic hardened-validator compromise ($p_c \leq 0.10$), Theorem 2 shows consensus reduces breach probability roughly $4\text{--}13\times$ over a single signature-verifying server; above the $p_c \approx 0.26$ crossover a hardened single verifier is preferable, so the case for consensus is real but bounded. Consensus earns its cost in multi-agency settings (e.g., cross-jurisdictional wildfire response with no single mutually trusted server), where Byzantine agreement over independently operated validators is the minimal trust model. The VIIRS-derived events preserve the qualitative ranking with only modest latency and F_p degradation (appendix), motivating field-scale validation before stronger claims.

Limitations Five limitations bear on the safety case. First, Theorem 1 is conditional on Assumption 1: the chaincode is not formally verified, so the guarantee holds for the specification, not a certified deployment; applying existing verification machinery (Bhargavan et al. 2016) is the highest-priority future work. Second, consensus is *simulated*: the Ed25519 signing, key authorization, nonce ledger, and threshold logic are real and exercised by the reported attacks, but BFT ordering is a Python delay model and no Fabric network was deployed, so consensus-layer latency and throughput under real network conditions remain unmeasured. Third, the fire model is a cellular automaton rather than a validated operational simulator (FARSITE/Phoenix), so absolute latencies should be read as within-model comparisons, not field-calibrated values; likewise F_p is a within-model false-discovery ratio inflated by repeat alerts on a spreading fire. Fourth, the policy-level comparators are surrogates reproducing soft-constraint enforcement semantics rather than reimplementations of the published algorithms (Section 6.1),

so they bound the regime rather than benchmark those methods. Fifth, the human-operator model is a stationary Bernoulli approval (Eq. (7)): the cryptographic guarantee is independent of operator error and the sensitivity sweep covers p_{err} up to 0.20 (appendix), but the model omits correlated fatigue and adversarial social engineering, the residual threat the cryptography deliberately does not address, since GOMDP secures *authorization integrity*, not *decision correctness*.

8 Conclusion

We introduced the Governance-Invariant MDP: safety constraints cryptographically enforced at the environment level, with a formally scoped safety case (Theorems 1–2) supporting the paper’s central contribution: a layered empirical decomposition for wildfire early warning that isolates what authentication, consensus, coordination, and verification each buy (Section 7). The path to deployment is staged (Section 8), from the present simulation on real fire distributions toward supervised field operation; formal chaincode verification, validated fire-simulator integration, and a correlated-error operator model are the technical prerequisites, and the framework extends to other safety-critical domains.

Broader Impact and Path to Deployment

GOMDP addresses a civic need, public wildfire warning, where false alarms and missed detections both carry real social cost. Its contribution is *accountability*: every public-alert decision is provably authorized by a named human operator and permanently, tamper-evidently logged, an audit trail current pipelines lack. Deployment is staged and reaches only stage one here (simulation on real fire distributions); hardware-in-the-loop trials with fire-agency operators and supervised field operation remain future work. UAV monitoring raises airspace and privacy concerns, and all components add no offensive capability.

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Table 3: Principal notation ($\tau=\tau_2=0.80$ throughout).

Symbol	Definition
GOMDP	Governance-Invariant MDP
$\mathcal{G}, \mathcal{T}_G$	Governance predicate; constrained transition
s_t	System state: $(H_t, W_t, D_t, \mathcal{R}_t)$
$H_t, W_t, D_t, \mathcal{R}_t$	Heat, weather, UAV config, risk estimate
b_t	Belief state $P(s_t \mid o_{1:t}, a_{1:t-1})$
$\mathcal{U}, N$	UAV fleet; fleet size
$\tau_1, \tau_2 = \tau$	Stage-1, stage-2/alert threshold ($= 0.80$)
$\text{Conf}_t^{(1)}, \text{Conf}_t^{(2)}$	Stage-1 and stage-2 confidence scores
HA_t	Human authorization binary variable
L_d, L_e	Detection latency; end-to-end latency
F_p, FN_r	False public alert rate; miss (false-negative) rate
P_{break}	Governance breach probability
p_c, k, f	Compromise prob.; validator count; tolerance
$p_{\text{spoofer}}, p_{\text{drop}}$	Sensor spoofing rate; packet drop rate
ϵ_{sig}	Ed25519 signature forgery probability

Table 4: Experimental Configuration Parameters.

Parameter	Value
Grid size	100×100
Timesteps / episode (T)	3000 (10 s each)
Random seeds (simulation)	20 (seeds 0–19)
UAV fleet sizes (N)	$\{5, 10, 20, 40\}$
Thermal detection probability	0.85
Fire-spread wts. ($\alpha_1, \alpha_2, \alpha_3$)	(0.45, 0.35, 0.30)
Blockchain platform	Hyperledger Fabric
Consensus	BFT ordering
Validators k / tolerance f	$7/2$ ($k \geq 3f+1$)
BC delay (nominal)	$\mathcal{N}(1.2, 0.3)$ steps
Human review delay	$\mathcal{N}(3.0, 0.8)$ steps
HITL approval	$\text{Bern}(1-p_{\text{err}})$, $p_{\text{err}}=0.05$
Thresholds ($\tau_1, \tau_2 = \tau$)	(0.60, 0.80)
Fusion weights (w_H, w_W)	(0.65, 0.35)
Cost weights (α, β, γ)	(0.50, 0.35, 0.15)
PPO learning rate	3×10^{-4}
PPO training episodes	1000
PPO hidden dims	(256, 128)
PPO observation normalization	Min-max per episode
PPO reward scaling	Max return, first 50 ep.
WCSAC safety budget	$\mathbb{E}[c] \leq 0.10$ (as PPO-CMDP)

A Notation and Configuration Parameters

Table 3 summarizes the principal notation and Table 4 lists all configuration parameters. Figures 3 and 2 show the fleet-size dependence of latency and false alert rate, Table 5 the full metric comparison at $N = 20$, and Table 6 the full spoofing and injection results.

B Full Proof of Theorem 1

We proceed by reduction. Suppose an adversary $\mathcal{A}$ produces an unauthorized alert ($\text{Alert}_t = 1$ despite $\mathcal{G}(s_t, a_t) = 0$). Under Assumption 1, $\mathcal{T}_G$ blocks any alert action without a valid governance certificate exactly as Definition 1 specifies, so $\mathcal{A}$ must achieve this through the certificate mechanism itself rather than through a faulty implementation. There are two remaining avenues: $\mathcal{A}$ must either (Case 1) produce a valid certificate without possessing an authorized validator key, or (Case 2) compromise more than f validators to achieve malicious consensus (addressed by Theorem 2). For Case 1, we construct an Ed25519 forger $\mathcal{B}$ from $\mathcal{A}$: given a public key pk and challenge message m (the anomaly event record), $\mathcal{B}$

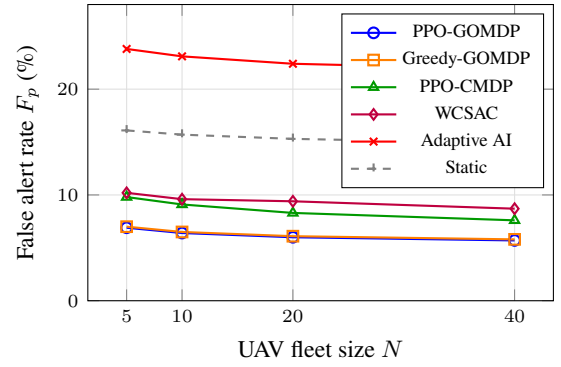


Figure 2: False public alert rate F_p vs. fleet size N under synthetic anomaly injection ($n=20$ seeds). Holm–Bonferroni-corrected $p < 0.01$ for all GOMDP vs. non-GOMDP comparisons.

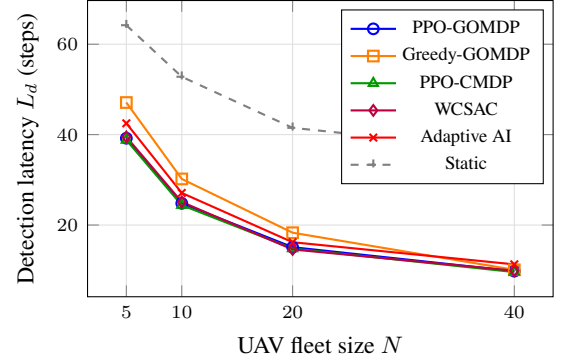


Figure 3: Detection latency L_d vs. UAV fleet size N ($n=20$ seeds); error bars are ± 1 SD. The Static curve is consistent with the Tarigan et al. (Tarigan, Robet, and Tarigan 2026) IoT figure (≈ 45 steps; context only, non-comparable).

runs $\mathcal{A}$ to produce a forged signature σ' and outputs (σ', m) . Since $\mathcal{A}$ succeeds in Case 1 with probability p_A , $\mathcal{B}$ breaks the $(t_{\text{adv}}, \epsilon_{\text{sig}})$ -security of Ed25519 with the same probability: $p_A \leq \epsilon_{\text{sig}}$. Applying a union bound over T timesteps yields Eq. (1). For practical parameters ($T = 3000$, $\epsilon_{\text{sig}} = 2^{-128}$), $T \cdot \epsilon_{\text{sig}} \approx 0$. We note explicitly what this proof does and does not establish: it does not rule out a third failure mode, a chaincode implementation bug that causes the deployed $\mathcal{T}_G$ to diverge from Definition 1 despite a valid certificate and an honest validator majority. That failure mode is excluded by hypothesis (Assumption 1), not by the cryptographic or consensus argument; Eq. (1) is therefore a guarantee about the specification conditional on faithful implementation, not an unconditional guarantee about any deployed system.

C Operational Guarantees

The following operational guarantees hold under the assumptions of Section 4. **Enforced Authorization:** $P(\exists t \leq T : \text{Alert}_t = 1 \wedge (\text{Conf}_t^{(2)} \leq \tau \vee \text{HA}_t = 0)) \leq T \cdot \epsilon_{\text{sig}} \approx 0$ for all policies, conditional on Assumption 1 (direct consequence of Theorem 1). **Alert Integrity:** Alert injection by an unauthorized actor is computationally infeasible under Ed25519 $(t_{\text{adv}}, \epsilon_{\text{sig}})$ -security and Byzantine tolerance $f \leq \lfloor (k-1)/3 \rfloor$

Table 5: Full metric comparison ($N=20$, $n=20$ seeds). FN_r : missed-detection rate; overhead relative to ungoverned Adaptive AI. ^{def}: compliance is definitional (cf. Table 1), not independently measured.

Metric	PPO-GOMDP	Greedy-GOMDP	PPO-CMDP	WCSAC	Adapt. AI	Static
L_d (steps)	15.1±1.1	18.3±1.4	14.8±1.0	14.6±1.2	16.2±1.2	41.5±3.1
F_p (%)	6.0±1.1	6.1±1.3	8.3±2.4	9.4±2.0	22.4±2.1	15.3±2.4
FN_r (%)	2.1±0.9	2.3±1.0	2.6±1.0	3.8±1.4	0.9±0.5	1.8±0.8
BC delay (steps)	1.2±0.3	1.2±0.3	—	—	—	—
Human rev. (steps)	3.0±0.8	3.0±0.8	3.0±0.8	—	—	—
Gov. compliance	100% ^{def}	100% ^{def}	92.8%	90.6%	0% ^{def}	0% ^{def}
Gov. overhead	−6.8%	+13.0%	−8.6%	−9.9%	baseline	—

Table 6: Adversarial robustness ($N=20$, 20 seeds): F_p (%) per attack; “inj.” rows report injection success. Central = Adaptive AI without authentication.

Attack / Condition	Param.	GOMDP	Cent.+Sig	Central
No attack	—	6.0	6.0	22.4
Spoofing (i.i.d.)	$p=0.05$	6.7	6.7	26.8
Spoofing (i.i.d.)	$p=0.10$	7.8	7.9	31.2
Spoofing (i.i.d.)	$p=0.20$	9.4	9.5	38.7
Spoofing (strategic)	$p=0.10$	8.6	8.7	34.5
Alert injection (success)	$p_{att}=1$	0/100	0/100	100/100

(Theorem 2). **Non-Repudiation and Replay Resistance:** All events and authorizations are signed, hashed, and permanently committed; per-event nonces prevent replays. These hold independently of the policy’s approximation quality (Corollary 1) and under Byzantine faults within tolerance.

D VIIRS-Data Simulation: Full Table

Table 7 tabulates the full per-event VIIRS-data results discussed in Section 6.3; all summary ranges are stated in the main-body prose.

Table 7: VIIRS-data evaluation ($N=20$, $n=20$ seeds); latency in simulation timesteps. [‡]: Tarigan et al. (Tarigan, Robet, and Tarigan 2026) is qualitative context only (non-comparable deployment); no improvement is claimed.

Event	Method	L_d	F_p (%)	Gov.
California ’20	PPO-GOMDP	22.4±3.2	8.3±2.1	100%
	Greedy-GOMDP	26.9±3.8	8.5±2.3	100%
	PPO-CMDP	22.0±3.1	10.6±2.7	93.1%
	Adaptive AI	20.1±2.9	24.6±3.8	0%
Mediterr. ’21	PPO-GOMDP	24.1±4.1	9.1±2.5	100%
	Greedy-GOMDP	28.8±4.6	9.3±2.6	100%
	PPO-CMDP	23.6±3.9	11.4±3.0	92.4%
	Adaptive AI	21.7±3.5	26.1±4.2	0%
NSW ’19–20	PPO-GOMDP	21.8±2.7	7.9±1.9	100%
	Greedy-GOMDP	26.1±3.3	8.2±2.1	100%
	PPO-CMDP	21.3±2.8	10.1±2.4	93.6%
	Adaptive AI	19.8±2.6	23.9±3.5	0%
—	IoT threshold [‡]	≈45	—	—

E Validator/Verifier Compromise: Full Table

Table 8 tabulates the stochastic-theory and deterministic-empirical validator/verifier compromise results discussed in Section 6.5 and Section 7; all values are stated in the main-body prose.

Table 8: Validator/verifier compromise ($k=7$, $f=2$). Theory: exact binomial tail of Eq. (2) (GOMDP) and p_c (single verifier). Empirical: f_c fixed compromised validators; breach iff $f_c \geq f+1=3$.

Stochastic (theory)			Deterministic (empirical)		
p_c	P_{break}^{GOMDP}	P_{break}^{sig}	f_c	Breach	F_p (%)
0.05	0.004	0.050	0	0/100	6.0
0.10	0.026	0.100	1	0/100	6.1
0.20	0.148	0.200	2	0/100	6.2
0.30	0.353	0.300	3	100/100	8.9

F Validator-Count Sweep

Table 9 tests Theorem 2’s scaling in k at fixed stochastic compromise probability $p_c = 0.10$, with tolerance $f = \lfloor (k-1)/3 \rfloor$. The theoretical column is the exact binomial tail of Eq. (2).

Table 9: Validator-count sweep at $p_c = 0.10$.

k	f	P_{break}^{GOMDP} (theory)	Empirical
4	1	0.052	0.054
7	2	0.026	0.025
10	3	0.013	0.013
13	4	0.006	0.006

G Threshold and HITL Sensitivity

F_p , L_d , and FN_r were evaluated over $\tau_1 \in \{0.50, 0.55, 0.60, 0.65, 0.70\}$ and $\tau_2 \in \{0.75, 0.80, 0.85, 0.90\}$. Governance compliance remains 100% across the entire grid for PPO-GOMDP and Greedy-GOMDP (Theorem 1 is threshold-independent). HITL approval volume decreases with τ_2 , raising FN_r by up to 1.4 pp at $\tau_2 = 0.90$; the chosen configuration ($\tau_1 = 0.60$, $\tau_2 = 0.80$) achieves the best precision-recall balance on the validation seed set, and qualitative rankings are preserved across the grid. Table 10 reports sensitivity to the operator error rate p_{err} : compliance remains 100% for all values (the cryptographic invariant is independent of operator error), FN_r increases monotonically with p_{err} as more genuine events are rejected, and F_p decreases slightly as stricter rejection also filters borderline false positives. This tradeoff underscores the importance of calibrating operator rejection in real deployments; a multi-signature (m -of- n) scheme is a candidate mitigation we have not implemented.

Table 10: HITL error-rate sensitivity ($N=20$, $n=20$ seeds, PPO-GOMDP); compliance is 100% throughout. Main results use $p_{\text{err}}=0.05$ (bold).

p_{err}	FN_r (%)	F_p (%)	Gov. Compliance
0.05 (main)	2.1 ± 0.9	6.0 ± 1.1	100.0%
0.10	3.9 ± 1.2	5.5 ± 1.0	100.0%
0.15	5.8 ± 1.5	5.1 ± 0.9	100.0%
0.20	7.8 ± 1.8	4.2 ± 0.8	100.0%

H m -of- n Multisignature Ablation

Section 6.4 identifies a lightweight m -of- n multi-signature quorum as a plausible, lower-overhead alternative to the full Hyperledger Fabric deployment. The design replaces the BFT consensus layer with m -of- n threshold signatures collected from n designated verifiers, requiring no blockchain ledger; non-repudiation is retained (each verifier’s signature is attributable) but on-chain immutability and Byzantine agreement across a larger validator set are not.

Table 11: m -of- n multisignature ablation ($N=20$, 20 seeds).

Configuration	L_d	F_p (%)	Inj.
m -of- n multisig	15.0 ± 1.1	6.1 ± 1.2	100/100

The values match Central+Sig’s injection resistance (same underlying authentication primitive: threshold signatures) while forgoing full BFT validator-compromise resilience, making it a candidate for deployments whose threat model tolerates a smaller trust quorum than seven-validator consensus.